Group Declaration

Introduction to Artificial Intelligence

Course 02180

Authors

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1 Contribution Declaration

The work for this project was carried out collaboratively by all group members. The contributions reflect both the implementation work and the corresponding sections in the report, and are summarized below:

1.1 Implementation Contributions

- **Michele Berlato**
  - Core game mechanics implementation.
  - Board and state representation design.
  - Integration of the search framework and game progression logic.
- **Marco Cola**
  - Graphs implementation (plot generation and visualization).
  - Move-generating procedure implementation.
  - Performance optimization and refactoring for efficiency.
- **Alexandros Kouros**
  - Minimax and Alpha-Beta search logic implementation.
  - Integration of heuristic evaluation within search algorithms.
  - Testing of algorithmic configurations and edge cases.
- **Mattia Russo**
  - User interaction layer and command-line interface.
  - Evaluation function structure and parameterization.
  - Development of benchmarking pipelines and automated match simulations.

1.2 Benchmarking and Evaluation

- **Mattia Russo**
  - Terminal-based benchmarking system.
  - Execution of benchmarking experiments.
  - Computation and collection of performance statistics.
- **Marco Cola**
  - Implementation and generation of graphical benchmarks (plots).
  - Interpretation and commentary on figures and experimental results and trends.

1.3 Report Writing Contributions

- **Section 1 - Game Rules:** Michele Berlato
  - Detailed description of Kalaha rules and gameplay mechanics.
  - Explanation of board representation and action semantics.
- **Section 2 - Game Type:** Alexandros Kouros
  - Formalization of the game type and structural properties.
  - Analysis of its suitability for adversarial search methods.

- **Section 3 - Size space:** Marco Cola
  - Derivation of upper bounds for the state space using combinatorial analysis.
  - Discussion of branching factor and computational complexity.
  - Analysis of why exhaustive search is computationally infeasible.
- **Section 4 - Elements of the game:** Marco Cola
  - Formal definition of the game as a search problem.
  - Specification of core components: initial state, player function, actions, result function, terminal test, and utility.
  - Mapping of theoretical concepts to the implemented game model.
- **Section 5 - AI states and moves representation:** Mattia Russo
  - Representation of the game state as a fixed-size vector encoding the board configuration.
  - Formalization of move representation as pit indices and their mapping to game actions.
  - Description of deterministic state transitions and their role in search.
- **Section 6 - Methods & Algorithms:** Michele Berlato
  - Explanation of Minimax and Alpha-Beta pruning algorithms.
  - Description of recursive evaluation and pruning conditions.
  - Construction of illustrative examples of algorithm execution.
- **Section 7 - Heuristics and Evaluation functions:** Mattia Russo
  - Evaluation function design and interpretation of heuristic terms.
- **Section 8 - Implemented AI's parameters:** Marco Cola
  - Definition of configurable AI parameters (depth and algorithm choice).
  - Generation and presentation of benchmark figures.
  - Analysis and interpretation of performance trends (time, win rate, node count).
- **Section 9 - Future work:** Mattia Russo
  - Identification of potential improvements and extensions.

1.4 Additional Project Materials

- **README File:** Marco Cola
- **Code Documentation and Inline Comments:** All group members
- **Repository setup and structure:** Marco Cola, Mattia Russo
- **Version control management (commits, branching, merging):** Marco Cola, Mattia Russo, Michele Berlato
- **Testing and Debugging Support:** All group members
- **Experiment Reproducibility and Code Organization:** Marco Cola, Mattia Russo, Alexandros Kouros

2 Final Approval

All authors reviewed, contributed to, and approved the final project and report.